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# Intelligent prediction of compressive strength of concrete based on CNN-BiLSTM-MA

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ABSTRACT

Various factors affect the compressive strength of concrete, and performing experiments on this is both time-consuming and costly. Constructing predictive models for the strength of concrete at different ages can significantly enhance the efficiency of testing and serve as a foundation for engineering quality analysis. Currently, concrete strength prediction using traditional machine learning models exhibits suboptimal accuracy, while the performance enhancement achieved by single deep learning models is limited. This study utilizes 8 variables associated with concrete strength as inputs for the predictive model, focusing on the compressive strength of concrete at different ages. In this study, we propose a hybrid model, referred to as CNN-BiLSTM-MA, which combines convolutional neural network (CNN), bidirectional long short-term memory neural network (BiLSTM), and multi-head attention mechanism (MA). The CNN module is employed to extract data feature relationships, while the BiLSTM module captures deep features within the concrete compressive strength data. Additionally, the MA module is utilized to focus on important features influencing concrete compressive strength, further improving the model's predictive accuracy. To determine optimal parameters for the model, five mainstream machine learning models, including Random Forest, Adaboost, GBDT, XGBoost, and LightGBM, were tuned using a random search algorithm and 5-fold cross-validation. Subsequently, these models were contrasted with our proposed model using the experimental dataset. The results indicate that the CNN-BiLSTM-MA model achieved the best predictive performance and generalization with RMSE = 1.21, MSE = 1.57, MAE = 0.85, and $R^2$ = 0.97.

## 1. Introduction

Due to the advantages of good plasticity and high strength, concrete is the most important load bearing material in the project where it is located, and it has been widely used in a variety of construction applications. Ensuring precise estimations of concrete compressive capacity is crucial during construction to uphold progress and quality. However, the forecast of concrete strength is a complicated task because of the various factors that determine the compressive capacity [1]. Concrete strength is often estimated with empirical calculations and laboratory testing in conventional methods. Refs. [2–4], however, there are some limitations in the above

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methods, such as the poor prediction of concrete strength for concrete with complex compositions, as well as the use of standardized concrete cube compressive strength tests that demand significant time and resources.

Over the past few years, the implementation of ML in the optimization of cement material mixtures and the prediction of strength performance has garnered extensive attention from researchers [5]. Many scholars have employed ML models for forecasting the concrete strength. Sun et al. [6] presented a approach using a weighted SVM combined with a fast chaotic genetic algorithm, using a dataset with 8 input variables and 1 output variable, achieving an RMSE of 10.401. Ahmad et al. [7] estimated the concrete strength using three traditional machine learning models, decision tree, Bagging regression, and AdaBoost, where the Bagging achieving superior performance compared to the other two, showing MSE = 3.75, $R^2$= 0.97, and MAE = 1.51. Qi et al. [8] applied Particle Swarm Optimization tuned Random Forest to the concrete compressive strength dataset, achieving testing set performance metrics of MSE = 27.535, MAE = 3.746, R = 0.954 and Explained Variance Score (EVS) = 0.901. Khan et al. [9] used XGBoost algorithm and after using vector multi-objective optimization parameters, achieved an $R^2$ of 0.949. Li et al. [10] investigated ensemble learning algorithms in HPC strength prediction by creating a new dataset that combined experimental data with previously published data and found that among the ensemble learning models, the GBDT model outperformed others, achieving an MAE of 0.139 and an RMSE of 0.182. In addition, various methods including linear regression [11,12], KNN [13], M5P [14], extra tree [15], genetic weighted pyramid operation tree(GWPOT) [16], extreme learning machine (ELM) [17], surrogate modeling techniques [18], LightGBM [19], group method of data handling [20,21], and GEP [22,23] have been applied to various concrete strength datasets, all of which have made some progress. Although certain machine learning algorithms have achieved promising results through intelligent hyperparameter tuning, concrete strength is influenced by multiple factors such as cement content, admixture dosage, water, and curing age, etc. The weak intercorrelation between these variables, coupled with the limitations of traditional machine learning models in capturing nonlinear relationships, may significantly compromise predictive performance when dealing with a substantial amount of feature input variables.

With the continuous development of artificial intelligence [24–26], shallow and deep neural networks have become increasingly effective instruments for estimating the concrete strength [27–30]. In contrast to conventional ML methods, which require manual selection and construction of features, neural network models are able to automatically extract higher-level, abstract features from data, thereby diminishing the necessity for feature engineering, and can better capture the complexity of concrete materials [31]. By comparing a multiple linear regression model with a BP neural network, Tam et al. [32] were able to estimate the compressive capacity of $CO_2$ concrete with greater precision. In a study by Imran et al. [33] on the prediction of HPC strength, 2171 experimental samples were gathered from existing literature and trained using the cascade forward neural network (CFNN), which was optimized with the artificial bee colony (ABC) algorithm, achieving $R^2$, RMSE, and MAE values of 0.953, 4.04, and 3.10, respectively. Using calcium carbide residue and rice husk ash (RHA) as cement substitutes, Jiang et al. [34] developed a novel that incorporates an Emotional Neural Network, which is optimized using Chaotic Particle Swarm Optimization model to investigate the strength-related characteristics of self-compacting concrete (SCC). Kumar et al. [35] obtained a dataset from the literature comprising 144 experiments with eight input parameters and employed MARS, deep neural networks (DNN), and extreme learning machines (ELM) to estimate concrete strength, with results indicating that the DNN performed best. Zeng et al. [36] introduced a deep CNN, which was trained on a dataset composed of 380 concrete mixtures and compared the prepared experimental dataset with AdaBoost, ANN and SVM, to validate the precision and reliability of their model in forecasting the concrete strength.

However, existing research on concrete strength prediction using deep learning models has the following shortcomings: 1)some studies apply simple neural network models, including DNN [37,38], CNN [39], LSTM [40], etc., which may not capture the complex features influencing concrete compressive strength adequately, resulting in limited improvements in predictive performance. 2)Most intelligent optimization algorithms have numerous hyperparameters, and their parameter configurations typically require expertise and domain-specific knowledge. Furthermore, finding the optimal settings could be resource-intensive and computationally demanding, increasing the intricacy of algorithm usage. 3)Single deep models often require significant data to achieve good performance. If the available concrete compressive strength data is limited, a single deep learning model may not have enough information for accurate predictions. Hybrid deep learning models can enhance data efficiency by integrating information from traditional methods, thereby improving model performance on datasets.

Nowadays, several hybrid deep learning approaches have been introduced and applied in a wide range of industries, for instance, traffic information detection [41], photovoltaic/wind power prediction in smart grid applications [42], stock price prediction [43], wind speed prediction [44], oil well production foresting [45], short-term power load prediction [46], and prediction of immediate performance deterioration in fuel cell systems [47], etc., but research on hybrid deep models for forecasting concrete strength is still limited.

Therefore, this study introduces an innovative model for forecasting the compressive strength at different ages of concrete, utilizing a CNN-BiLSTM-MA approach to address the problems of low prediction accuracy, the need to incorporate complex intelligent optimization algorithms, and the lengthy tuning process that exist in the current traditional machine learning model for concrete strength prediction. A multilayer CNN module is responsible for extracting the feature relationships from all input parameters of the concrete dataset, and the BiLSTM neural network is responsible for capturing the deeper features in the concrete compressive strength data to strengthen the model's understanding of the nonlinear relationships of the data. Moreover, the model incorporates a multi-head attention mechanism, enabling it to simultaneously focus on different features of the input data, thereby better capturing both global and local relationships. Simultaneously, by setting Dropout layer, the model prevents the neural network from becoming overly reliant on specific features or neurons during the training process. This aids the model in adapting better to new, unseen data, mitigating overfitting, and ultimately achieving accurate predictions for concrete strength across different curing periods.

The main contributions of this study include:

(i) A novel hybrid deep learning model has been developed, integrating a convolutional neural network (CNN), bidirectional long short-term memory (BiLSTM), and a multi-head attention (MA) mechanism.
(ii) A dataset for concrete compressive strength was constructed based on multiple specific mix proportion schemes, containing 8 different input features, including water-binder ratio, binder, flyash, sand ratio, superplasticizer, air-entraining agent, cast concrete temperature and age.
(iii) An extensive performance evaluation was conducted using RMSE, MSE, MAE, and $R^2$ as metrics. Several renowned ML models were also employed for the current task, including RF, Adaboost, GBDT, XGBoost, and LightGBM.

## 2. Research framework

The overall approach of this research involves the integration of convolutional neural networks, bidirectional long short-term memory neural networks, and multi-head attention mechanisms to predict the compressive strength of concrete. The overall workflow diagram in Fig. 1 is shown with the outlined steps:

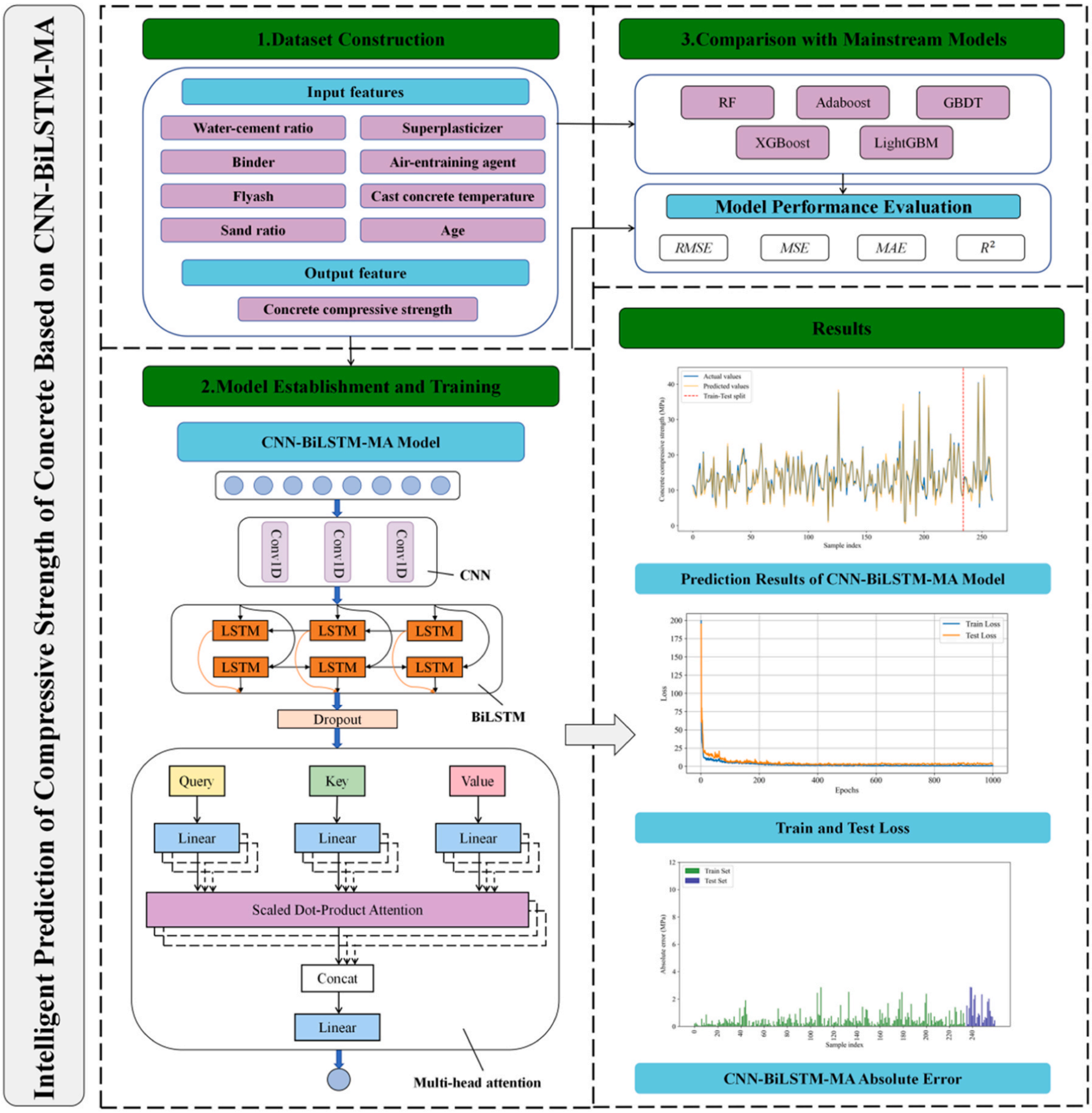


Fig. 1. Research framework.

(1) Data preprocessing: Concrete sample data from multiple specific mix proportion schemes, along with the constituent data for each concrete sample, were collected. The dataset was constructed with 8 input features and 1 output feature, representing the concrete's compressive strength. To ensure comparability and facilitate analysis, the data were standardized to eliminate scale differences among the variables.
(2) Model Training: We established the CNN-BiLSTM-MA model and conducted training and prediction on the concrete strength dataset. To evaluate the model's forecasting accuracy, four evaluation metrics were employed: RMSE, MSE, MAE, $R^2$.
(3) Model Comparison: Popular models currently employed for concrete strength prediction, including RF, Adaboost, GBDT, XGBoost, and LightGBM, were optimized using a grid search algorithm and assessed on both the training and test datasets through 5-fold cross-validation. When compared to the model presented in this study, the results highlight the precision and superiority of our model.
(4) Model Applicability Analysis: To further assess the model's robustness, additional experiments were conducted on datasets with single-factor variations, including Age_Variation and Temp_Variation, to examine its performance under different conditions.

## 3. Analysis of factors affecting

The compressive capacity of concrete is a critical measure impacted by a range of factors that interact with each other, collectively determining the ultimate performance of concrete. Dantas et al. [48] selected factors including water-cement ratio, mortar content, and cement content, achieving good results in predicting concrete strength using an artificial neural network. Zhang et al. [49] considered factors including superplasticizer, water, cement, water reducer, and fly ash and obtained field concrete test data and trained various machine learning models, achieving better predictive performance compared to previous research. Ke et al. [50] conducted a Sobol sensitivity analysis to investigate the main variables influencing the compressive capacity of concrete and revealed that among the three binders, fly ash (FA), cement, and blast furnace slag (BFS), FA played the most crucial role. Liu and Sun [51] implemented an interpretable boosting method to explore the contributions involving mix design factors to compressive capacity, global and local interpretability analyses indicated that curing age and cement content played a decisive role in determining concrete compressive strength, while water reducers had a positive influence on the predictive outcomes. Yi, Moon, and Kim's study [52] revealed that the curing temperature of concrete also affects its compressive strength, introducing temperature as a factor to enhance the prediction accuracy. Ma et al. [53] carried out investigations into the strength evolution of recycled and natural concrete in cold conditions, using exponential and logarithmic functions to forecast concrete strength, demonstrating the influence of temperature on its prediction. In order to comprehensively consider the factors influencing concrete compressive capacity and enhance the accuracy of model predictions, this study selected 10 parameters as input features. These parameters include water-binder ratio, binder content, fly ash, sand ratio, superplasticizer, air-entraining agent, cast concrete temperature and age. According to the characteristics of each parameter, the influencing factors were divided into three categories: mix design factors, curing conditions, and curing age.

## 4. Methodology

### 4.1. CNN

CNN is designed to efficiently process grid-like data as a deep learning model [54]. Fig. 2 illustrates the basic structure of CNN. In comparison to artificial neural networks (ANN), CNN's strength comes from its capacity to enhance feature extraction through greater network depth. [55]. Within the hybrid model, the CNN module comprises the following: (1) an input layer that receives selected variables as input; (2) The hidden layer primarily includes convolutional and fully connected layers. The convolutional layer is crucial for feature extraction, utilizing kernels to identify important patterns in the input data. With an increase in the quantity of kernels, the features extracted become progressively more abstract and refined. Because of the high number of data input parameters and complex relationships between input features in this study, three consecutive one-dimensional convolutional layers are used for feature extraction from the input data. In the convolutional layers, each neuron is connected only to a subset of neighboring neurons, minimizing network complexity and the amount of parameters. Additionally, the RuLU activation function is used to enhance the nonlinear representation of the CNN layer; (3) The output layer, responsible for passing the extracted features to the BiLSTM layer, performs this function seamlessly.

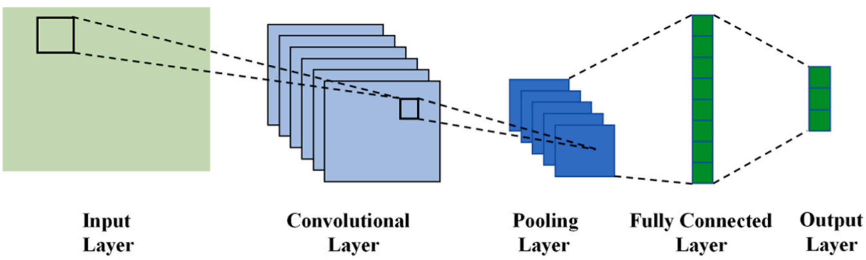


**Fig. 2.** Basic structure diagram of CNN.

CNN possesses the property of parameter sharing, allowing it to effectively handle data with translational invariance. However, this parameter sharing is not suitable for concrete compressive strength data because different positions in the data may require different feature extraction weights. In contrast to CNN, BiLSTM provides a more flexible approach to modeling data, unconstrained by local sliding windows. They can adaptively learn complex relationships between features, aiding in the discovery of more abstract and deeper features within concrete compressive strength data. This, in turn, enhances the accuracy of concrete compressive strength prediction.

### 4.2. BiLSTM

The problems of gradient vanishing and exploding in lengthy sequences are addressed by LSTM, a type of recurrent neural network that employs cells with gating mechanisms to control input flow [56]. The LSTM cells consist of input gates, cell states, forgetting gates, and output gates, each of which controls the movement of information through the network. This enables the LSTM to maintain information in sequence data for a long time and deliver information selectively [57].

BiLSTM introduces bi-directional processing based on LSTM. It enables information flow and capture in both directions of the input sequence by the model. It includes two independent hidden layers, each handling one direction, thereby improving its ability to retain long-range dependencies within the dataset [58]. The structure is shown in Fig. 3.

For concrete compressive strength data, different input features exhibit varying correlations and degrees of influence on concrete compressive strength. In order to allow the model to emphasize different features and their interrelations independent of position, Thus enhancing the model's understanding and predictive capacity, this study introduces a multi-head attention mechanism.

### 4.3. Multi-head attention mechanisms

The attention mechanism is a technique used to calculate the degree of correlation between various positions in an input sequence [59]. The core idea is to adjust the weights of input information to change the importance of different input information, thereby focusing on crucial information, suppressing irrelevant information, and ultimately improving the model's predictive accuracy.

The output attention matrix is computed as:

$$Attention\left(\boldsymbol{Q},\boldsymbol{K},\boldsymbol{V}\right) = softmax\left(\frac{\boldsymbol{Q}\boldsymbol{K}^{T}}{\sqrt{dk}}\right)\boldsymbol{V} \tag{1}$$

Where, $dk$ is the feature dimention of each key for weight scaling, normalized by softmax to the interval [0,1].

Multi-head attention is an extension of the self-attention mechanism that incorporates several distinct attention heads. Each head possesses a unique set of weight parameters for queries, keys, and values to capture different types of relations and features [60].

The multi-head attention mechanism splits the data series into $h$ subspaces, and each head performs a self-attention computation on the subspaces, which in turn enhances its representational capacity. Subsequently, The outputs from the $h$ heads are concatenated and integrated to obtain multiple heads. Concatenating these separate heads yields the final output, which is then transformed linearly.

$$headi = Attention(\boldsymbol{Q}\boldsymbol{W}_i^Q, \boldsymbol{K}\boldsymbol{W}_i^K, \boldsymbol{V}\boldsymbol{W}_i^V) \tag{2}$$

Where, $\boldsymbol{W}_i^Q$, $\boldsymbol{W}_i^K$, $\boldsymbol{W}_i^V$ are the weight matrices.

$$MultHead(\boldsymbol{Q},\boldsymbol{K},\boldsymbol{V}) = Concat(head1,\ldots,headh)\boldsymbol{W}^o \tag{3}$$

Where, $\boldsymbol{W}^o$ is the weight for linear transformation; *Concat* represents the splicing operation. Fig. 4 illustrates the model structure.

### 4.4. CNN-BiLSTM-MA model structure

The structure of the CNN-BiLSTM-MA-based model for predicting concrete compressive strength is shown in Fig. 5. The CNN-

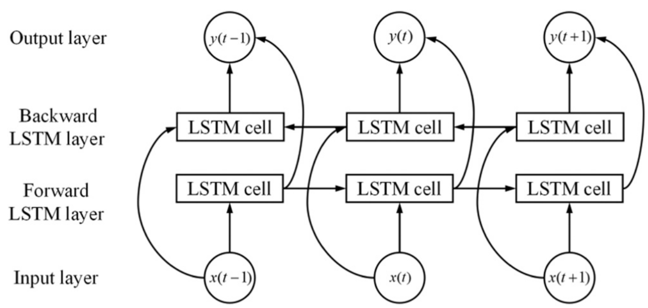


**Fig. 3.** BiLSTM structure diagram.

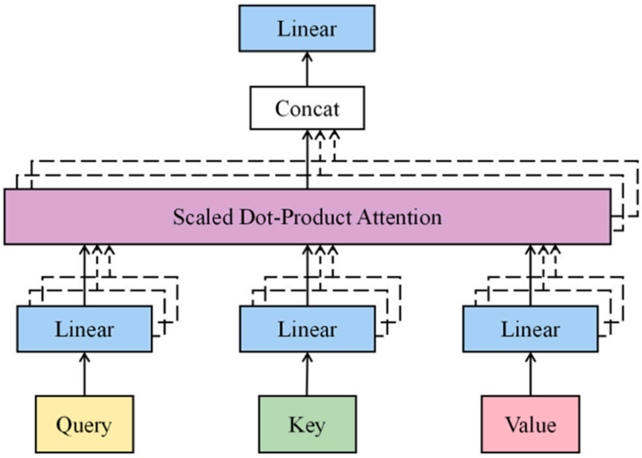


**Fig. 4.** Structure of the multi-head attention mechanism.

BiLSTM-MA model is composed of six parts: the input layer, the CNN module, the BiLSTM layer, the Dropout layer, the multi-attention layer and the output layer. The input concrete compressive strength sample data is received and normalized by the input layer. The CNN module is composed of one-dimensional convolutional layers, which extract feature information from the concrete compressive strength prediction data.Stacking multiple convolutional layers enhances the model's feature extraction capabilities. Employing wide convolutional kernels in the convolutional layers broadens the receptive field, enabling the efficient reduction of noise interference and the extraction of additional feature information. The sample data processed by the CNN module enters the BiLSTM layer. The BiLSTM layer leverages its bidirectional structure to capture deep and complex relationships between different features in concrete compressive strength data, thereby enhancing the model's understanding of nonlinear relationships within the data. After the BiLSTM layer, a Dropout layer is introduced, which randomly discards some of the neurons' outputs during training, thus lessening the overfitting of the model. This helps enhance the model's generalization capability, leading to better performance on the test data. The multi-head attention layer receives the output from the Dropout layer, allowing the hybrid model to learn to focus on various types of information in different attention heads which are then integrated together, enabling it to handle more complex relationships and dependencies without being influenced by the position of the data. As the final step, the output layer maps the output of the attention layer to a single output value for the concrete strength forecasting task. The above analysis demonstrates that, when compared with the

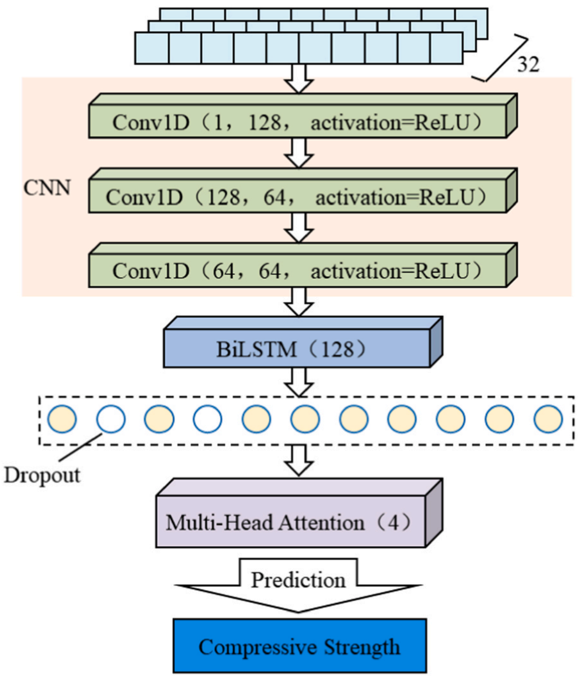


**Fig. 5.** Structure of the CNN-BiLSTM-MA model.

individual CNN and BiLSTM models, the hybrid model in our study not only extracts hidden features from the data and emphasizes key factors influencing concrete compressive strength, but also captures internal data correlations, leading to improved prediction performance.

### 4.5. Evaluation metrics

In this study, four metrics are applied to assess the performance of the prediction model: root mean square error (RMSE), mean square error (MSE), mean absolute error (MAE) and coefficient of determination ($R^2$). $R^2$ indicates the linear correlation between predicted and observed results and takes values from 0 to 1. MSE measures the mean squared error, while RMSE, as its square root, reflects how much predictions deviate from actual values. Similarly, MAE evaluates a model's accuracy by averaging the absolute differences between observed and predicted values. The prediction approach performs better when the MSE, RMSE, and MAE values are smaller. The expressions for these four performance matrics are as follows:

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n}\left(y^{obs} - y^{pred}\right)^2}{n}} \tag{4}$$

$$MSE = \frac{\sum_{i=1}^{n}\left(y^{obs} - y^{pred}\right)^2}{n} \tag{5}$$

$$MAE = \frac{1}{N}\sum_{i=1}^{n}\left|y^{obs} - y^{pred}\right| \tag{6}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(y^{obs} - y^{pred}\right)^2}{\sum_{i=1}^{n}\left(y^{obs} - \overline{y}^{obs}\right)^2} \tag{7}$$

Where, $y^{obs}$ and $y^{pred}$ represent the observed and predicted values of concrete compressive strength, respectively.

## 5. Experiment and discussion

### 5.1. Data preprocessing

The data was obtained from a large-scale hydropower construction site, based on specific mix proportion parameters that cover the majority of scenarios encountered in actual construction, providing a certain degree of generalizability. The input parameters include the water-binder ratio, binder content, fly ash content, sand ratio, superplasticizer content, air-entraining agent content, temperature of the cast concrete, and age. The output parameter is the compressive strength of the concrete, measured in MPa. A total of 271 data sets were collected, which were used to establish a dataset for predicting the compressive strength of concrete. Among them, 260 sets of data were divided into the training and testing sets for the model. The ranges of each influencing factor are listed in Table 1. Fig. 6 illustrates the correlations among the variables in the dataset. According to the correlation matrix, it can be observed that the linear relationships between most variables are not strong, suggesting that the hybrid model employed in this research is better at forecasting compressive strength. The model developed in this study is better suited to capturing non-linear connections among variables, which improves the accuracy of compressive strength forecasting.

Then the input features are normalized using Eq. (8). In this paper, Z-score standardization is used, which can effectively deal with features of different scales and make them have similar scales, while retaining the distribution shape of the original data and not changing the distribution characteristics of the data. It can also help the model converge faster and reduce the training time of the

**Table 1**
Statistics for the dataset variables.

| Variables | Type | Maximum | Minimum | Mean | Standard Deviation |
|---|---|---|---|---|---|
| Water-binder ratio | Input | 0.5 | 0.36 | 0.45 | 0.034 |
| Binder($kg/m^3$) | Input | 329 | 166 | 193.35 | 35.56 |
| Flyash(%) | Input | 35 | 0 | 34.45 | 3.833 |
| Sand ratio(%) | Input | 42 | 22 | 25.27 | 2.610 |
| Superplasticizer(%) | Input | 0.7 | 0.5 | 0.65 | 0.068 |
| Air-entraining agent(%) | Input | 0.05 | 0.005 | 0.04 | 0.006 |
| Cast concrete temperature(°C) | Input | 15.7 | 6.0 | 8.38 | 2.187 |
| Age(days) | Input | 28 | 1 | 4.11 | 2.755 |
| CC strength(MPa) | Output | 41.8 | 1.2 | 13.71 | 5.595 |

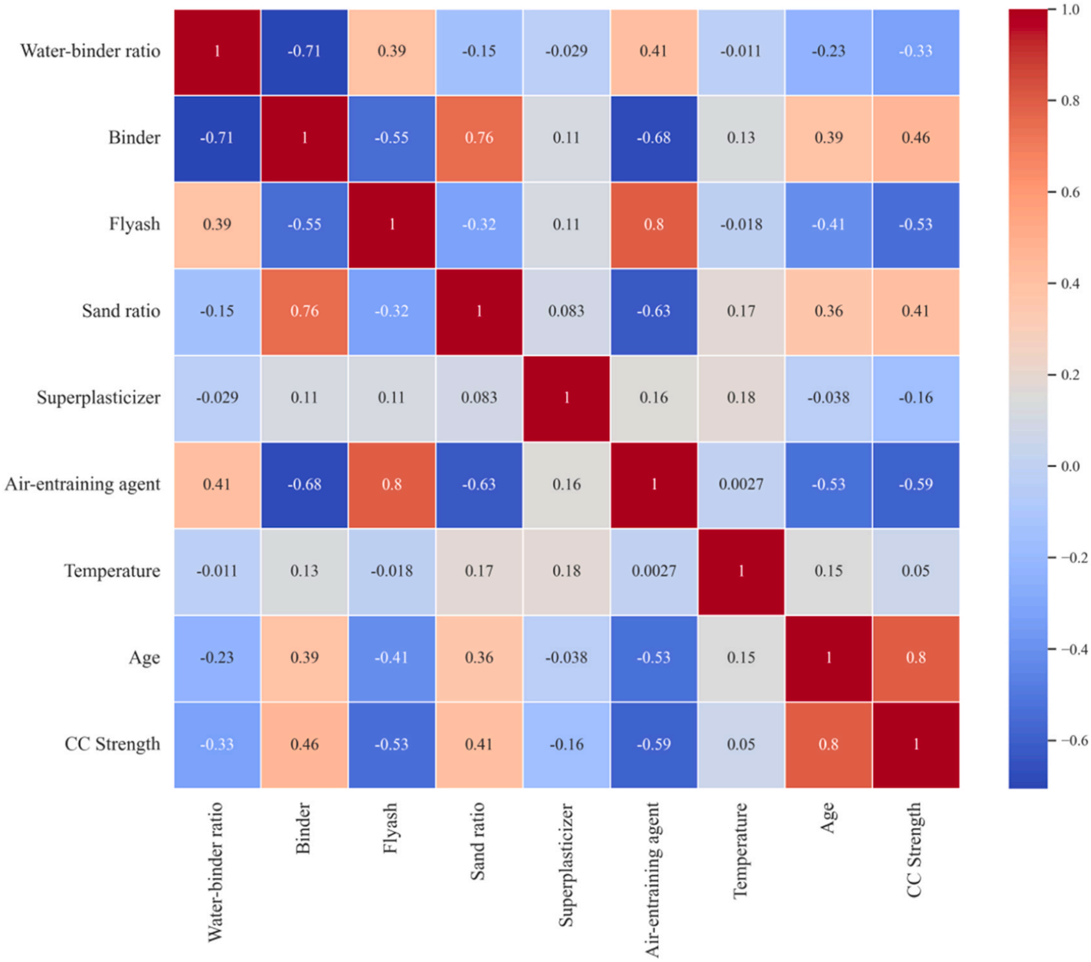


**Fig. 6.** Pearson coefficient correlation matrix.

model.

$$Z = \frac{X - \mu}{\sigma} \tag{8}$$

Where, X, μ and σ stand for the original value, mean, and standard deviation of the dataset, respectively.

### 5.2. CNN-BiLSTM-MA model evaluation

This section investigates the performance effect of CNN-BiLSTM-MA model on concrete compressive dataset. The dataset is partitioned randomly into a training set and a test set in a 9:1 proportion. Since the good performance of a deep learning model is mainly influenced by the appropriate range settings of its hyperparameters. Therefore, in this research, the hyperparameters of the created hybrid model in particular the quantity of neurons, optimizer, batch size, and number of iterations are adjusted iteratively to obtain a better model for forecasting concrete strength. The hybrid model used in this study comprises a CNN module, a BiLSTM module and a multi-head attention module. In this model, the CNN module consists of three consecutive one-dimensional convolutional neural networks with 3 × 3 convolution kernels. These kernels specify a stride of 1 for the one-dimensional convolution window. The ReLU activation function is used, as it is faster compared to the sigmoid function, providing significant benefits in terms of neural network training and reduced training time. BiLSTM module comprises two LSTM units, which receive input from the convolutional layer. The input features are represented by 100 dimensions, while the hidden state consists of 128 units, with each LSTM unit containing 128 neurons. This layer processes data in a batch-first manner, ensuring that computations at each time step are performed in parallel. The dropout of 0.65 follows the BiLSTM module. The multi-head attention module is configured with 4 attention heads and input dimensions that are twice the size of the hidden state dimension. In the actual experimental process, the Adam optimizer is applied to train the model for 1000 iterations at an initial learning rate of 0.001.

Fig. 7 shows the model's training and testing results for forecasting the compressive capacity of concrete, highlighting the capability of the created hybrid model. The x-axis represents the index of individual data points, while the y-axis indicates the compressive strength. The red dashed lines before and after are the fitting curves of the training and test sets, respectively, the blue lines are the observed values, and the orange lines are the predicted values of the hybrid model. Overall, the model exhibits excellent fitting performance on both the training and test sets. Fig. 8 depicts the changes in the loss function of the hybrid model as the number of epochs grows in the training set and test set phases. It is important to highlight that the loss function gradually stabilizes around 400 epochs and reaches very low loss values, indicating excellent predictive performance.

### 5.3. Comparison with mainstream machine learning models

To provide further evidence of the effectiveness of the CNN-BiLSTM-MA model, an evaluation was made with five mainstream ML models: RF, Adaboost, GBDT, XGBoost, and LightGBM. Model performance was comprehensively evaluated using four metrics: RMSE, MSE, MAE, and $R^2$. Following data standardization, 90 % of the data was utilized as the train set, with the remaining 10 % reserved for the test set. For each ensemble model, a random search algorithm was used together with 5-fold cross-validation to optimize their hyperparameters, ensuring that the models perform at their best. The optimal parameters for the compared models, obtained through the combination optimization algorithm, are detailed in Table 2. These parameters were subsequently employed in the respective models for experimentation. The forecasted outcomes of the hybrid model and the five ensemble models on the test set and training set are presented in Table 3. The created hybrid model demonstrates the best performance. In terms of the results on the test set, when comparing the CNN-BiLSTM-MA model with the RF model, RMSE, MSE, and MAE are reduced by 59.12 %, 82.32 %, and 59.13 %, respectively, while $R^2$ increases by 11.49 %. Compared to the created hybrid model and Adaboost model, RMSE, MSE, and MAE decrease by 66.76 %, 88.12 %, and 68.16 %, respectively, with an increase in $R^2$ by 19.75 %. In comparison with the CNN-BiLSTM-MA model and GBDT model, RMSE, MSE, and MAE decrease by 47.39 %, 70.38 %, and 48.80 %, respectively, while $R^2$ increases by 5.43 %. When comparing the hybrid model with the XGBoost model, RMSE, MSE, and MAE decrease by 48.73 %, 71.86 %, and 54.05 %, respectively, with an increase in $R^2$ by 5.43 %. Finally, in contrast with the CNN-BiLSTM-MA model and LightGBM model, RMSE, MSE, and MAE decrease by 57.99 %, 81.11 %, and 53.55 %, respectively, with an increase in $R^2$ by 10.23 %. The comparative results reveal that the created hybrid model, which combines CNN with BiLSTM, enhances the model's accuracy in forecasting concrete strength dataset. Fig. 9 illustrates the discrepancy curves of predicted values and observed values for each model on the test set. The red solid circles stand for the predicted values for each model, while the blue squares depict the observed values in the test set. In the figure, the vertical axis represents the compressive strength values, including both predicted and observed values, measured in MPa, while the horizontal axis corresponds to the index of individual data points. From the figure, it is clear that among the ensemble learning methods, XGBoost and GBDT models exhibit good fitting performance, with relatively small prediction errors for each data point and a trend that closely aligns with the actual trend. However, their performance significantly deteriorates when dealing with outlier points. In contrast, the CNN-BiLSTM-MA model not only displays minimal prediction errors for data points, with a trend that perfectly matches the actual trend but also effectively fits the outlier points within the data. This highlights the performance and robustness of the model proposed in this research.

For a more intuitive view of the forecasting ability of each model on the dataset, Figs. 10–15 provide the absolute error plots for the hybrid model and the comparison models. The x-axis represents the index of individual data points, with the first portion corresponding to the training set data and the latter part representing the test set data. The y-axis indicates the absolute error of the predicted values, which is calculated as the absolute difference between the predicted and observed values for each data point. From Figs. 10–14, it is evident that among the five comparative models, GBDT and XGBoost models perform the best. They exhibit a roughly similar distribution of absolute errors across the entire dataset, with relatively small prediction errors for most data points. Specifically,

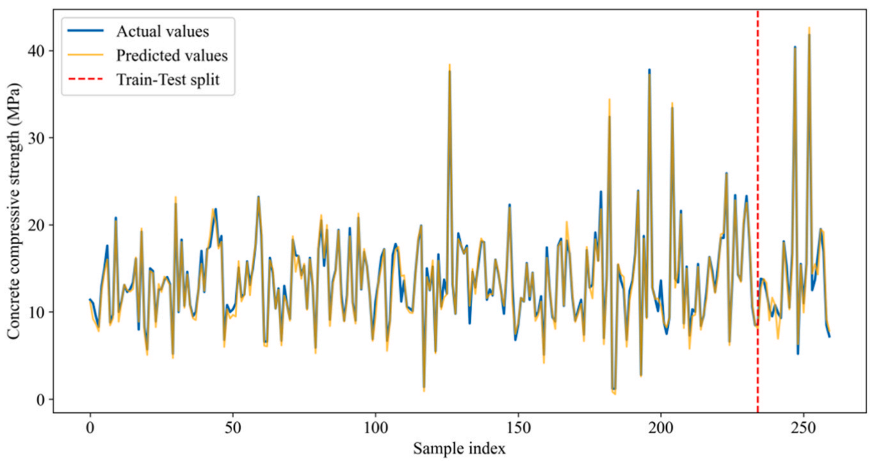


**Fig. 7.** Comparison curves of actual values and predicted values of CNN-BiLSTM-MA model for concrete compressive strength dataset. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

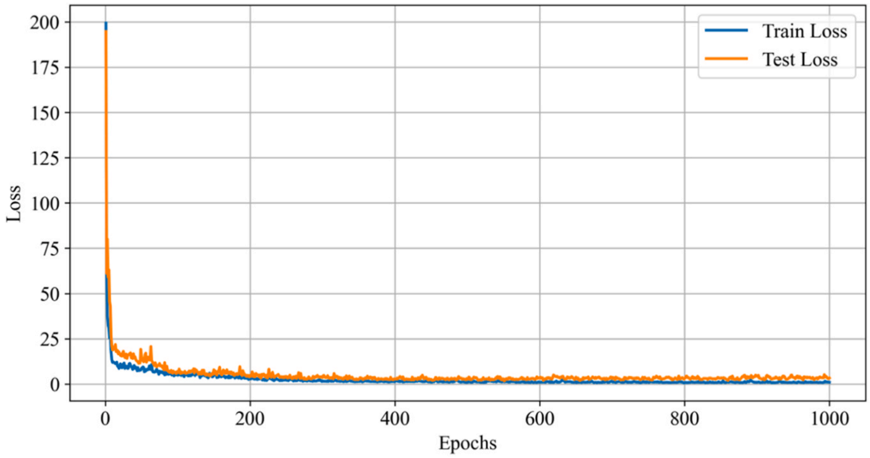


**Fig. 8.** Loss function profiles of CNN-BiLSTM-MA model at training and testing phases.

**Table 2**
Summary of hyperparameter optimization.

| Model | n_estimators | learning_rate | max_depth | Subsample |
|---|---|---|---|---|
| RF | 150 | _ | 10 | _ |
| Adaboost | 200 | 0.5 | _ | _ |
| GBDT | 350 | 0.1 | 3 | 0.5 |
| XGBoost | 300 | 0.2 | 3 | 0.3 |
| LightGBM | 200 | 0.7 | 3 | 0.6 |

**Table 3**
Comparison with mainstream ML models.

| Model | | RMSE | MSE | MAE | $R^2$ |
|---|---|---|---|---|---|
| CNN-BiLSTM-MA | Train Set | 0.65 | 0.37 | 0.32 | 0.99 |
| | Test Set | 1.21 | 1.57 | 0.85 | 0.97 |
| RF | Train Set | 1.53 | 1.82 | 1.18 | 0.95 |
| | Test Set | 2.96 | 8.78 | 2.08 | 0.87 |
| Adaboost | Train Set | 2.33 | 5.49 | 1.92 | 0.82 |
| | Test Set | 3.64 | 13.22 | 2.67 | 0.81 |
| GBDT | Train Set | 0.96 | 0.92 | 0.72 | 0.97 |
| | Test Set | 2.30 | 5.30 | 1.66 | 0.92 |
| XGBoost | Train Set | 1.13 | 1.05 | 0.75 | 0.96 |
| | Test Set | 2.36 | 5.58 | 1.85 | 0.92 |
| LightGBM | Train Set | 1.31 | 1.70 | 1.12 | 0.94 |
| | Test Set | 2.88 | 8.32 | 1.83 | 0.88 |

the maximum absolute error for XGBoost is 6.59 MPa, lower than the maximum absolute error of 7.69 MPa for the GBDT model. When compared with the proposed model and the XGBoost model, our model has an overall smoother distribution of absolute errors and achieves smaller errors for the majority of data points, as shown in Fig. 15. Its maximum absolute error is 2.85 MPa, which is also than the error observed in the XGBoost model. This indicates that the created model performs optimally on the dataset, demonstrating robustness against extreme values and exhibiting strong generalization performance.

### *5.4. Evaluation of model generalization ability*

To further validate the applicability of the model, the performance of the CNN-BiLSTM-MA model was investigated using data with single-factor variations. Eleven sample data points that were not involved in the previous training and testing phases were divided into two small datasets. Among them, six samples had age values gradually varying from 2 days to 7 days while keeping other parameters constant, referred to as Age_Variation. The remaining five samples had cast temperatures gradually changing from 6.2℃ to 6.6℃ while keeping other parameters unchanged, referred to as Temp_Variation. These two small datasets were separately fed into the established model for testing, and the RMSE, MSE, MAE, and $R^2$ metrics were obtained. The results of the CNN-BiLSTM-MA model on the Age_-Variation dataset are shown in Table 4, where the four evaluation metrics are generally consistent with the results on the test set. Fig. 16 presents a comparison between traditional machine learning models and the CNN-BiLSTM-MA model on the Age_Variation

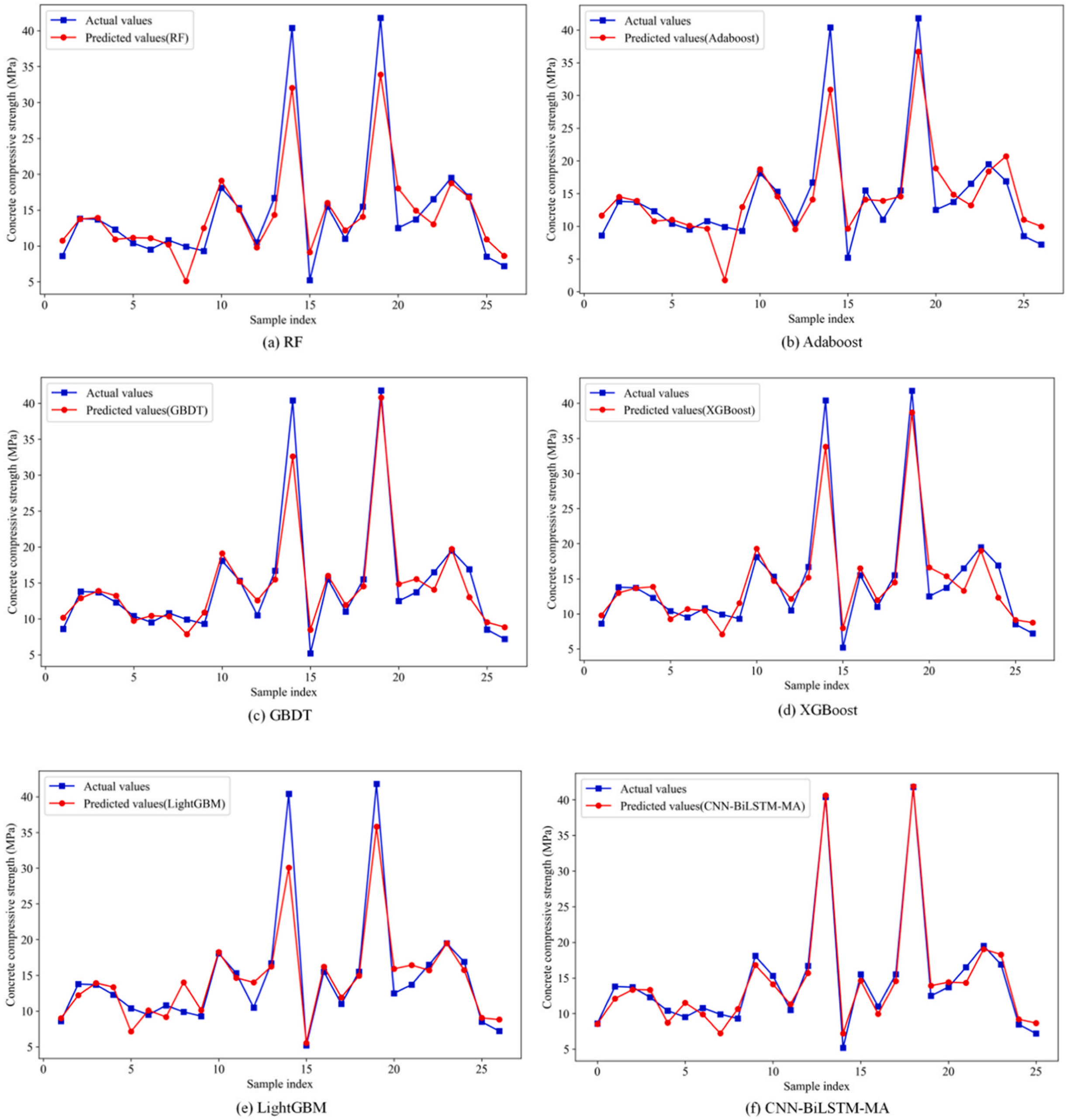


**Fig. 9.** Prediction resuluts of concrete compressive strength for each model on the test set. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

dataset. The results demonstrate that the hybrid model performs well on the Age_Variation dataset.

Table 5 presents the prediction performance of the CNN-BiLSTM-MA model on the Temp_Variation dataset. It can be observed that the RMSE, MSE, MAE, and $R^2$ values under single-factor cast temperature variations are similar to those obtained on the test set. This indicates that the model's predictive capability does not significantly degrade due to changes in this factor, demonstrating good generalization performance. Furthermore, Fig. 17 provides an intuitive comparison with traditional machine learning models, highlighting the advantages of the CNN-BiLSTM-MA model on the Temp_Variation dataset and verifying its applicability and stability under varying environmental parameters. In summary, the CNN-BiLSTM-MA model exhibits stable predictive performance on single-factor variation datasets, further confirming its strong applicability.

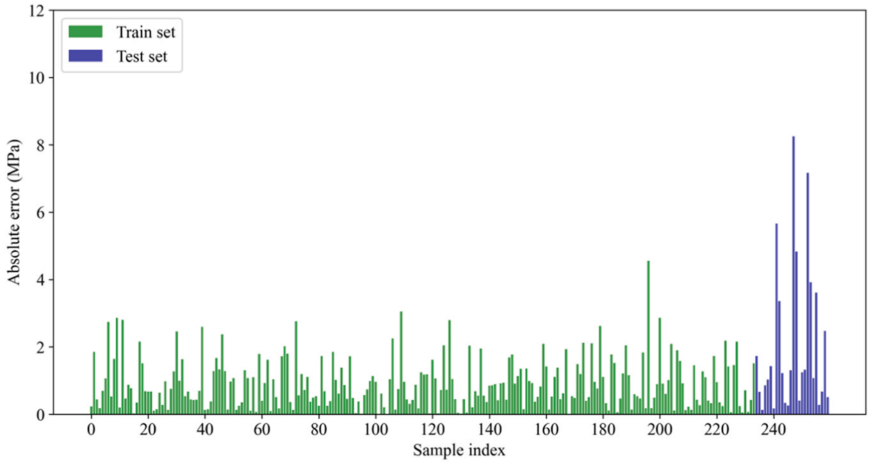


**Fig. 10.** Absolute error plot for RF model.

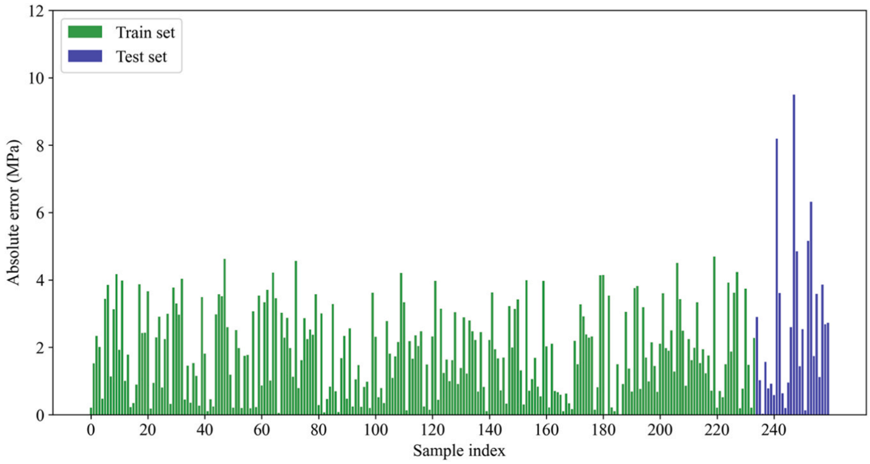


**Fig. 11.** Absolute error plot for Adaboost model.

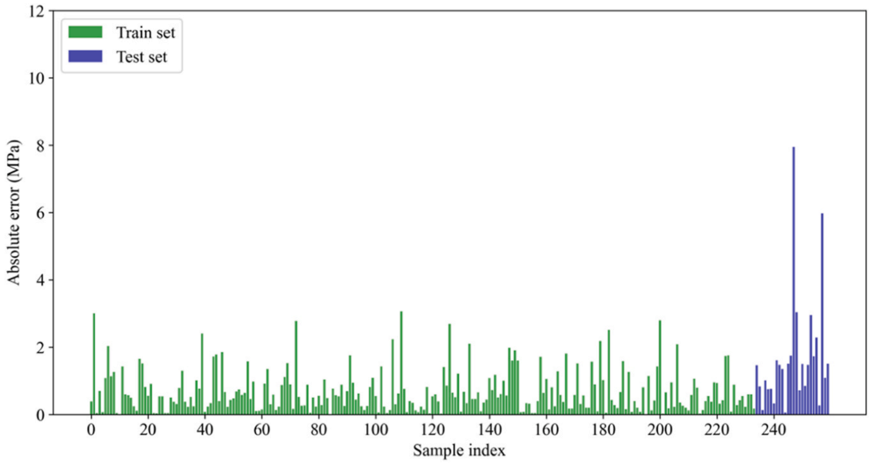


**Fig. 12.** Absolute error plot for GBDT model.

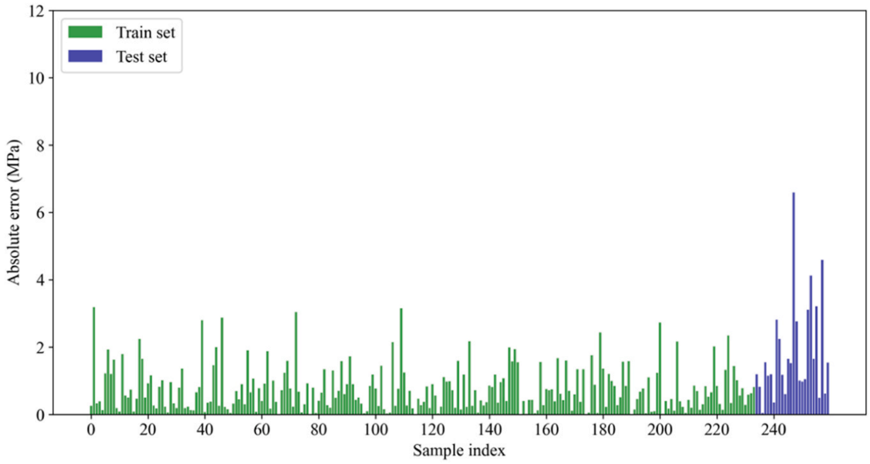


**Fig. 13.** Absolute error plot for XGBoost model.

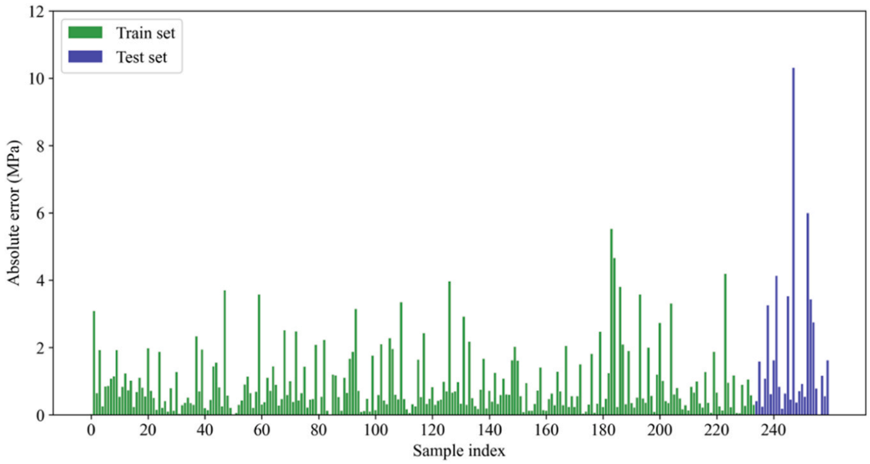


**Fig. 14.** Absolute error plot for LightGBM model.

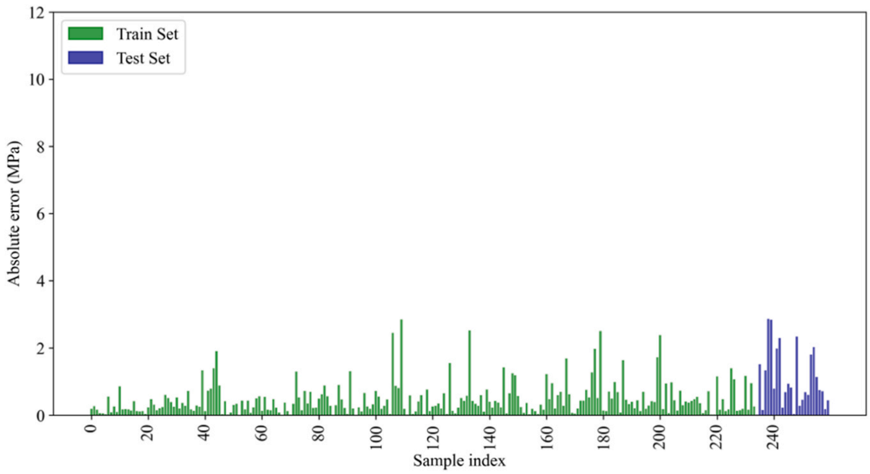


**Fig. 15.** Absolute error plot for CNN-BiLSTM-MA model.

**Table 4**
Performance evaluation of the CNN-BiLSTM-MA model on the Age_Variation dataset.

| Model | | RMSE | MSE | MAE | $R^2$ |
|---|---|---|---|---|---|
| CNN-BiLSTM-MA | Test Set | 1.21 | 1.57 | 0.85 | 0.975 |
| | Age_Variation | 1.31 | 1.66 | 0.88 | 0.973 |

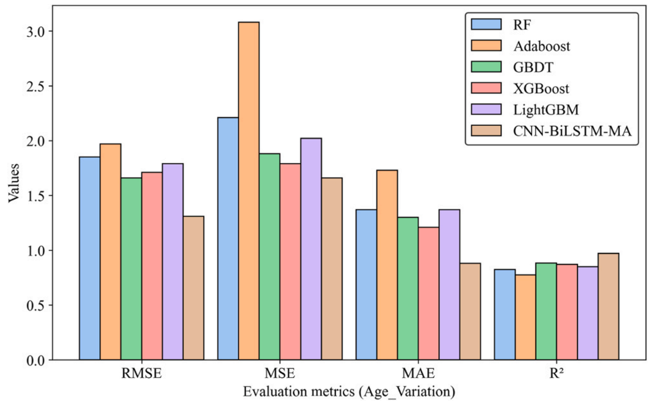


**Fig. 16.** Comparison of Prediction Performance of Different Models on the Age_Variation Dataset.

**Table 5**
Performance evaluation of the CNN-BiLSTM-MA model on the Temp_Variation dataset.

| Model | | RMSE | MSE | MAE | $R^2$ |
|---|---|---|---|---|---|
| CNN-BiLSTM-MA | Test Set | 1.21 | 1.57 | 0.85 | 0.975 |
| | Temp_Variation | 1.39 | 1.71 | 0.92 | 0.969 |

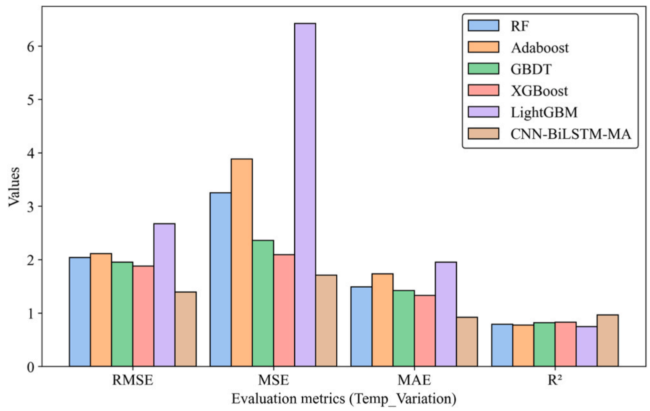


**Fig. 17.** Comparison of Prediction Performance of Different Models on the Temp_Variation Dataset.

## 6. Conclusions

Precise prediction of concrete compressive strength has important reference significance for the progress and quality control of construction projects. Concrete strength is influenced by numerous factors, and there exist highly nonlinear relationships among the input feature parameters. In light of the current limitations of traditional ML methods in precisely forecasting the compressive capacity of complex-component concrete, along with the inability of single deep learning models to adequately capture the intricate features affecting it, this study presents a model for prediction using CNN-BiLSTM-MA.

(1) This study obtained a dataset comprising 271 sets of concrete compressive strength data, encompassing 8 influencing factors, including water-binder ratio, binder, flyash, sand ratio, superplasticizer, air-entraining agent, cast concrete temperature and

age. Preliminary analysis was conducted on the fundamental information and correlations among various feature parameters of the dataset.

(2) An innovative CNN-BiLSTM-MA hybrid model structure was proposed. This model utilizes three consecutive one-dimensional convolutional neural networks to extract concrete data features deeply. BiLSTM is responsible for capturing the deep features in concrete compressive strength data, and precise predictions of concrete compressive strength are achieved through a multi-head attention mechanism. By adding a Dropout layer mitigates issues related to model overfitting. Evaluation metrics for prediction results were established. On the test set, the model achieved an RMSE of 1.21, MSE of 1.57, MAE of 0.85, and $R^2$ of 0.97. The discrepancy curves of predicted values and observed values on the test set demonstrate the model's high accuracy and strong generalization performance.

(3) Experiments were conducted on the same dataset using the current mainstream traditional machine learning models RF, GBDT, XGBoost, and LightGBM, and the hyper-parameters of the models were tuned by a approach of random search algorithm and 5-fold cross-validation. The experimental outcomes demonstrate that the hybrid model introduced in this research outperforms the best mainstream model, achieving a 5.43 % higher $R^2$ and a 56.8 % reduction in absolute error. These results demonstrate the effectiveness of integrating CNN, BiLSTM, and multi-head attention in predicting concrete compressive strength, surpassing traditional ML models.

(4) To further evaluate the model's applicability, additional experiments were conducted on two independent datasets with single-factor variations: Age_Variation and Temp_Variation. The results show that the CNN-BiLSTM-MA model maintains stable predictive performance under different conditions, with minimal deviations from the test set results. Comparative analysis with traditional ML models further confirms its superior adaptability and robustness, demonstrating strong generalization ability for practical applications.

## CRediT authorship contribution statement

**Purui Chen:** Writing – review & editing, Visualization, Supervision. **Tao Guan:** Writing – review & editing, Supervision, Resources, Funding acquisition. **Hongling Yu:** Writing – review & editing, Supervision, Resources. **Yuqiao Liu:** Writing – original draft, Visualization, Validation, Methodology, Data curation, Conceptualization. **Bingyu Ren:** Writing – review & editing, Supervision, Funding acquisition. **Zhenbang Guo:** Writing – review & editing.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data availability

Data will be made available on request.

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